

Intro to Blazing Web Accessibility (*)

(*) Empowered with AI

2021 // Dennie Declercq // @DennieDeclercq

Techorama

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Introduction

Who am I?

- Dennie Declercq
- DDSOft
- President / Developer
- @DennieDeclercq
- #DreamingIsBelieving



Why?

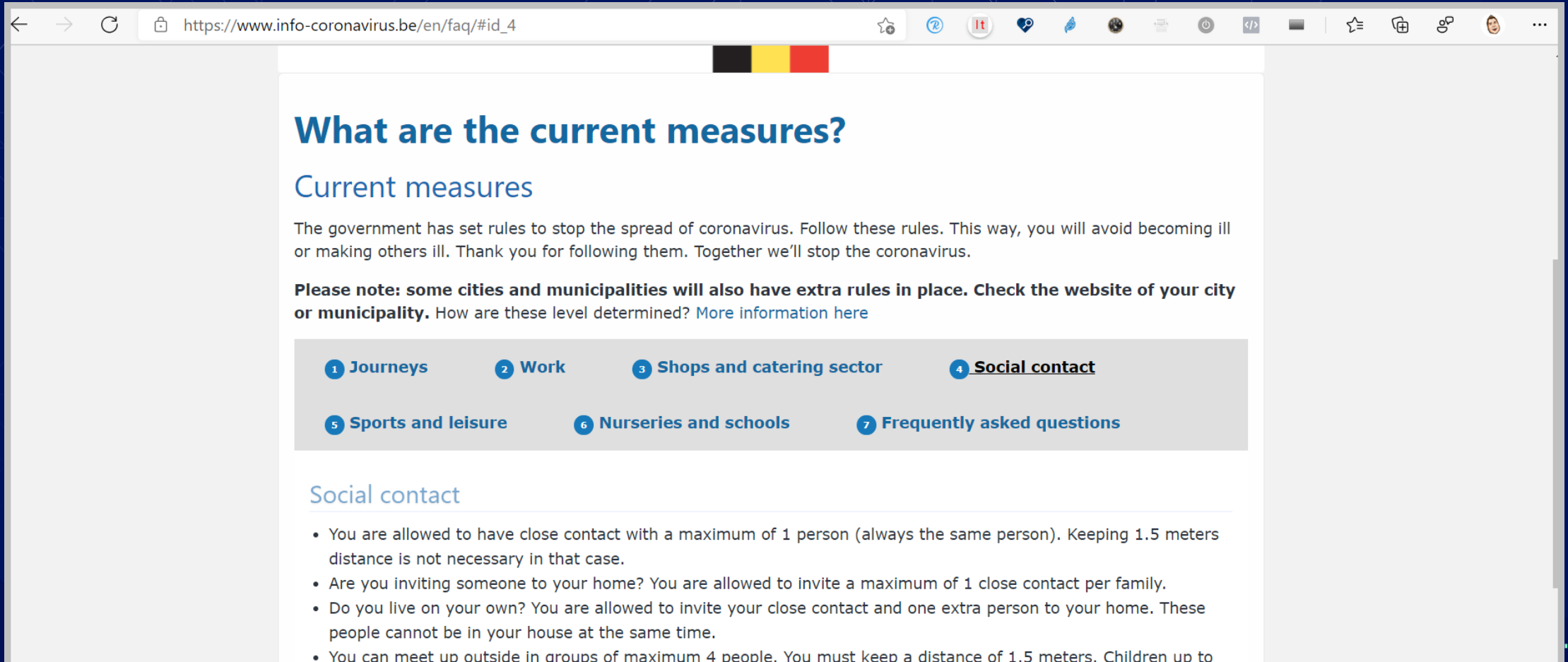
Blazor

- Never 'had JS under control'
- Context switching is hard (Autism)

Cognitive Services

- Possibilities that weren't possible
- True horse power

2021: COVID19 – Inaccessibility - Exclusion



The screenshot shows a web browser window with the URL https://www.info-coronavirus.be/en/faq/#id_4. The page title is "What are the current measures?". Below the title, the text states: "The government has set rules to stop the spread of coronavirus. Follow these rules. This way, you will avoid becoming ill or making others ill. Thank you for following them. Together we'll stop the coronavirus." A note follows: "Please note: some cities and municipalities will also have extra rules in place. Check the website of your city or municipality. How are these level determined? [More information here](#)". A horizontal menu contains seven items: 1 Journeys, 2 Work, 3 Shops and catering sector, 4 Social contact (highlighted), 5 Sports and leisure, 6 Nurseries and schools, and 7 Frequently asked questions. The "Social contact" section is expanded, showing a list of rules: "You are allowed to have close contact with a maximum of 1 person (always the same person). Keeping 1.5 meters distance is not necessary in that case.", "Are you inviting someone to your home? You are allowed to invite a maximum of 1 close contact per family.", "Do you live on your own? You are allowed to invite your close contact and one extra person to your home. These people cannot be in your house at the same time.", and "You can meet up outside in groups of maximum 4 people. You must keep a distance of 1.5 meters. Children up to".

What are the current measures?

Current measures

The government has set rules to stop the spread of coronavirus. Follow these rules. This way, you will avoid becoming ill or making others ill. Thank you for following them. Together we'll stop the coronavirus.

Please note: some cities and municipalities will also have extra rules in place. Check the website of your city or municipality. How are these level determined? [More information here](#)

- 1 Journeys
- 2 Work
- 3 Shops and catering sector
- 4 **Social contact**
- 5 Sports and leisure
- 6 Nurseries and schools
- 7 Frequently asked questions

Social contact

- You are allowed to have close contact with a maximum of 1 person (always the same person). Keeping 1.5 meters distance is not necessary in that case.
- Are you inviting someone to your home? You are allowed to invite a maximum of 1 close contact per family.
- Do you live on your own? You are allowed to invite your close contact and one extra person to your home. These people cannot be in your house at the same time.
- You can meet up outside in groups of maximum 4 people. You must keep a distance of 1.5 meters. Children up to

DEMO

Intro to Accessibility

Difficulties

- Learning Disability
 - Reading
 - Understanding / Cognition
 - Structure
- Autism / Autism Spectrum Disorder
 - Structure
 - Confirmation
- Dyslexia
 - Reading
- Mental Health
 - Structure (sometimes)
 - Confirmation

Oh yes...



- And the well known Accessibility too
 - Blindness or vision loss
 - Auditory disability
 - Motoric restrictions

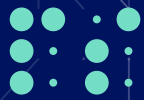
The Power of Text-To-Speech in Mobile apps



No, No, No Narrator or Voice Over

Narrator / Voice Over

- Blind / Vision Loss



- How does it work?
- ➔ Reads sequentially whole screen

Text-To-Speech

- Learning Disability
- Dyslexia



- How does it work?
- ➔ Click a control and read aloud

Intro to Web Accessibility

The POUR Principle

- Perceivable
- Operatable
- Understandable
- Robust

Semantic HTML

- Modern semantic HTML 5 is the start
- Correct use of headings
- Correct use of html 'Lang' attribute
- Sections
- Divs

Structure

- Simple navigation menu
- No – or – less submenu's
- Descriptive menu labels
- Logical separation

ALT Text for Images

- Really important for screen readers

Forms

- Be aware of timers
 - Can cause stress
 - Can cause panic attacks
- Label each Form Element

Layout guidelines

- Be aware of Color Contrast
- Be consistent
- Never use only one mark for (important) actions
 - Pictograph & Color
 - Underline & Color
 - Image & Text

Intro to Blazor

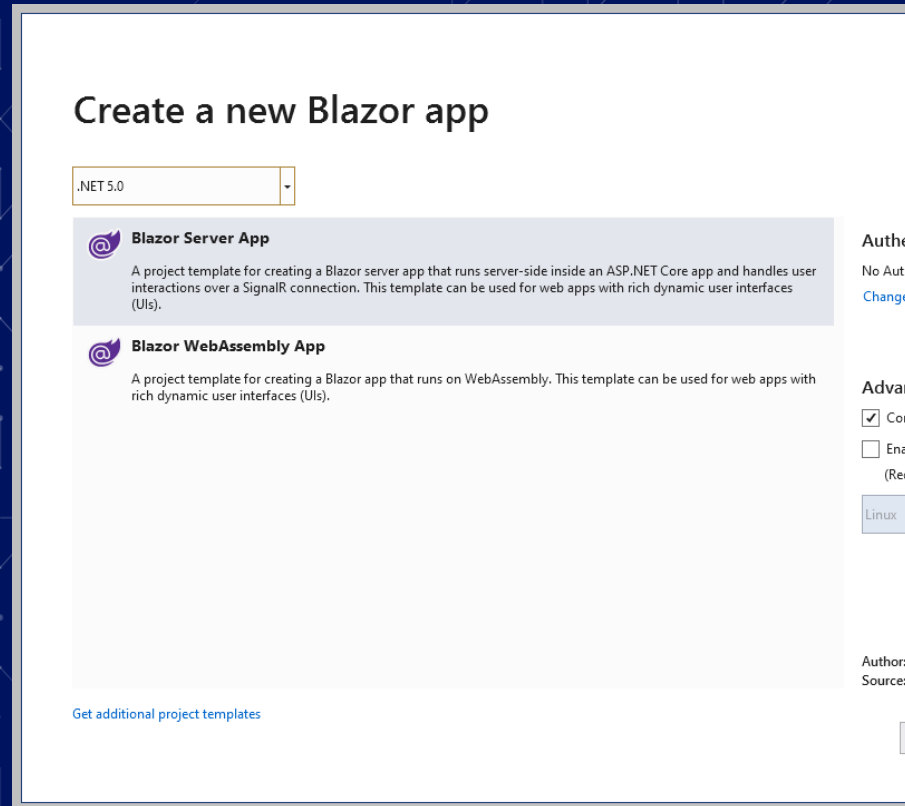


What is Blazor?

- Microsoft's Brand New PWA/ Frontend language
- 2 Flavors:
 - Blazor Server Side (Hosted as .NET/ .NETCore Web App)
 - Blazor Web Assembly
- Affordable solution:
 - Azure Storage with Azure Static Web Apps

Getting started with Blazor

- VS 2019
- VS Code



Exploring the Razor Syntax

```
page "/counter"

<h1>Counter</h1>

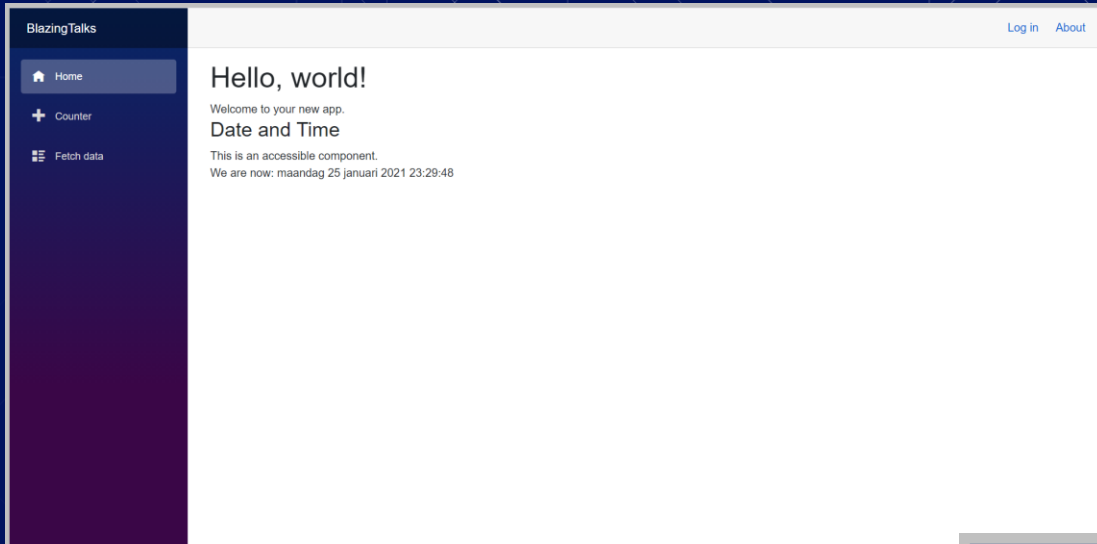
<p>Current count: @currentCount</p>

<button class="btn btn-primary" @onclick="IncrementCount">Click me</button>

@code {
    private int currentCount = 0;

    private void IncrementCount()
    {
        currentCount++;
    }
}
```

Razor components



```
@page "/"

<h1>Hello, world!</h1>

Welcome to your new app.

<AccessibleComponent />
```

```
<h3>Date and Time</h3>
<div>
  <p>
    This is an accessible component. <br />
    We are now: @DateTimeString
  </p>
  @code {
    string DateTimeString = DateTime.Today.ToLongDateString() + " " + DateTime.UtcNow.ToLongTimeString();
  }
</div>
```

DEMO: Intro to Blazor

Intro to Cognitive Speech

What is Microsoft Cognitive Services?

- Microsoft's AI platform
- Azure
- Pay as you go
- Developer focused
- API Calls

SOME SERVICES:

AI + Machine Learning > Cognitive Services

 Bing Autosuggest Microsoft Help users complete queries faster by adding intelligent type-ahead capabilities	 Bing Custom Search Microsoft Create a custom search engine with ad-free results	 Bing Entity Search Microsoft Recognize and classify named entities, and find search results based on them	 Bing Search Microsoft Enable safe, ad-free search for your users with Bing, a leading search engine	 Bing Spell Check Microsoft Help users identify and fix word breaks, slang, names, homonyms, and brands	 Computer Vision Microsoft An AI service that analyzes content in images	 Content Moderator Microsoft An AI service that detects unwanted content
 Custom Vision Microsoft An AI service and end-to-end platform for applying computer vision to your specific scenario	 Face Microsoft An AI service that analyzes faces in images	 Speaker Recognition Microsoft An AI service that enables you to identify individual speakers or use speech as a means of verification	 Speech Microsoft An AI service that enables you to embed human-like speech capabilities into your apps	 QnA Maker Microsoft An AI service that enables you to create a bot from your data in minutes	 Text Analytics Microsoft An AI service that enables you to unlock insights from natural language text using sentiment	 Translator Text Microsoft An AI service that enables you to easily translate text between more than 60 languages
 Personalizer Microsoft An AI service that delivers a personalized user experience	 Language Understanding Microsoft Teach your apps to understand commands from your users	 Ink Recognizer (Preview) Microsoft An AI service that recognizes digital ink content, such as handwriting, shapes, and document layout	 Immersive Reader (preview) Microsoft An AI service that helps users read and comprehend text	 Form Recognizer (preview) Microsoft The AI service that extracts information from documents	 Anomaly Detector (preview) Microsoft An AI service that helps you foresee problems before they occur	 Cognitive Services Microsoft Connect powerful AI to your apps

DEMO: Creating a Cognitive Service on Azure

Speech Service

- Text-to-speech
- Custom Voice



DEMO: Speech API

DEMO: The real stuff

Step-By-Step Guide

Microsoft Cognitive Speech



 Microsoft
Cognitive Services

Creating a Speech Service

[Home](#) > [New](#) > [Marketplace](#) >

Speech

Microsoft



Speech

Microsoft

[Save for later](#)

Create

Overview

Plans

Usage Information + Support

Transcribe audible speech into readable, searchable text. Add real-time speech to your apps and services using the programming languages you a

NuGet Package

Microsoft.CognitiveServices.Speech

Browse

Installed

Updates

Search (Ctrl+L)



Include prerelease



Microsoft.CognitiveServices.Speech by Microsoft

Microsoft Cognitive Services Speech SDK

Microsoft.CognitiveServices.Speech
Microsoft Cognitive Services Speech SDK

Keys.cs

Public Static Class with connection keys to Microsoft Cognitive Services

0 references

```
public static class KEYSDEMO
{
    public static string Key1 = "";
    public static string Region = "northeurope";
}
```

CSSpeechService.cs

- Code To call speech API

```
public async Task SynthesisToSpeakerAsync(string text)
{
    // Creates an instance of a speech config with specified subscription key and service region.
    // Replace with your own subscription key and service region (e.g., "westus").
    var config = Microsoft.CognitiveServices.Speech.SpeechConfig.FromSubscription(KEYS.Key1, KEYS.Region);

    // Creates a speech synthesizer using the default speaker as audio output.
    using (var synthesizer = new SpeechSynthesizer(config))
    {
        using (var result = await synthesizer.SpeakTextAsync(text))
        {
            if (result.Reason == ResultReason.SynthesizingAudioCompleted)
            {
                I
```

JavaScript in Host.cshtml

- Select HTML element:
 - Heading
 - Paragraph
 - ...

```
<script>  
    window.getElementText = (element, text)  
    {  
        text = element.innerText;  
        return text;  
    }  
</script>
```

PageCode



```
graph LR; A["@inject IJSRuntime"] --> B["+ next Slides"]
```

@inject
IJSRuntime

+ next
Slides

Adding References to Elements

- Elements:
 - Headings
 - Paragraphs
 - ..
- Add: `@Ref="RefName" <EL @ref="RefName">Content</EL>`
- In `@Code` block:
 - Private ElementReference RefName

```
<h1 @ref="PageHead">Passion</h1>
```

```
private ElementReference PageHead;
```

ReadAloud method

```
private async Task<bool> ReadAloud(ElementReference el  
    infoFromJs = await JSRuntime.InvokeAsync<string>(      
        "getElementText", elementRef);      
    await speech.SynthesisToSpeakerAsync(infoFromJs);      
    return true;
```

EventHandlers

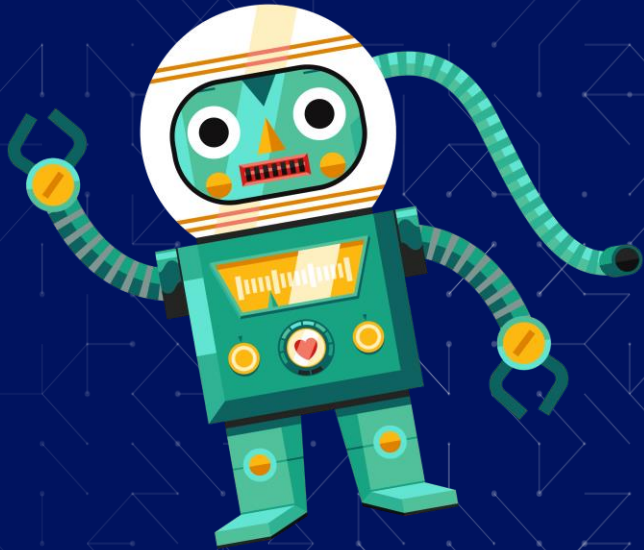
```
private async Task<bool> ReadHead()  
{  
    await ReadAloud(PageHead);  
    return true;  
}
```

Closing – Q&A

Thanks!

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- @DennieDeclercq
- #DreamingIsBelieving





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